MouseHash Tool Target Bundle Map

Mapping 100 finding-structure tools to the six-role MouseHash bundle: Conditions, Stimuli, Behavior, Neural Data, Time Organization, Metadata

Purpose

This document turns the 100-tool MouseHash map into a target-signature reference. Each tool is described by the bundle roles it needs, the role subcategories it targets, and the kind of tensor/view it usually materializes. The goal is not to force all neuroscience experiments into one rectangular array. The goal is to give every tool a typed scientific target so an agent can decide whether a dataset has the right ingredients before running the tool.

Core bundle roles

Role	Subcategories / meaning
Conditions	Task labels, trial labels, stimulus labels, experimental groups, brain states, session phases, task epochs.
Stimuli	Sensory stimuli, model-derived features, semantic features, interventions, stimulation, stimulus embeddings.
Behavior	Choices, reaction times, running, pupil, pose, kinematics, licking, reward, behavioral state.
Neural data	Spikes, calcium events/traces, LFP, EEG/ECoG, population matrices, latent states, fitted neural responses.
Time organization	Continuous time, trials, epochs, frames, events, lags, windows, alignments, trajectory segmentation.
Metadata	Subject/session, brain region, probe/electrode/unit/cell identity, acquisition, preprocessing, model/tool provenance.

How to read a target signature

Required means the tool is scientifically under-specified without that role. **Optional** means the role can improve stratification, annotation, leakage control, interpretation, or reporting. **Target subcategories** tells the agent what to look for inside each role. **Typical view** describes the array/table/query shape most often materialized from the role bundle.

Family-level target summary

Family	Dominant role tuple	Common targets
Encoding / decoding	Encoding: (Stimuli or Behavior, Neural Data, Time). Decoding: (Neural Data, Conditions or Behavior, Time).	Stimulus-response models, behavioral covariates, class labels, train/test-safe prediction.
Dimensionality reduction / factor modeling	Mostly (Neural Data), often plus (Conditions, Time, Metadata).	Population structure, latent factors, trajectories, session/area comparisons.
Geometry / representational structure	(Neural Data) plus one comparison axis: Stimuli, Conditions, Behavior, or Metadata.	RSMs, semantic axes, subspaces, manifolds, alignment across models/sessions/areas.
Dynamics / state-space	(Neural Data, Time), often plus Conditions/Behavior/Metadata.	Temporal evolution, hidden states, attractors, lagged influence, trajectories.
Connectivity / interaction	(Neural Data, Time, Metadata), often plus Conditions.	Unit/area interactions, correlations, coupling, assemblies, functional graphs.
Statistics / validation / visualization	The target roles of the tested or plotted tool, plus Time and Metadata for correct resampling/reporting.	Nulls, uncertainty, leakage controls, multiple comparisons, plots, auditable reports.

Encoding / decoding

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
1	Fit Linear Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Stimuli: feature matrix, semantic/model features; Behavior: continuous covariates; Neural: response vector/matrix; Time: aligned observations/windows	observations x features -> observations x neurons
2	Fit Ridge Encoding Model	Required: Stimuli, Neural data, Time organization Optional: Behavior, Conditions, Metadata	Stimuli: high-dimensional ViT/CLIP/PCA features; Neural: spike/calcium response matrix; Time: stimulus-response alignment; Metadata: neuron/session grouping	observations x high-d features -> responses
3	Fit Lasso Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Stimuli/Behavior: candidate sparse predictors; Neural: single-neuron or population responses; Time: aligned bins/trials	observations x features -> neuron response
4	Fit Elastic Net Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Stimuli/Behavior: correlated predictors; Neural: responses; Time: observation index	observations x features -> responses
5	Fit Poisson GLM Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Neural: spike counts/event counts; Stimuli/Behavior: covariates; Time: count windows, lags; Conditions: trial labels	time/trial bins x predictors -> count
6	Fit Negative Binomial Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Neural: overdispersed counts; Stimuli/Behavior: predictors; Time: bins/windows; Metadata: neuron/session	bins x predictors -> overdispersed count
7	Fit Zero-Inflated Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Neural: sparse events/zero-heavy counts; Stimuli/Behavior: predictors; Time: trials/bins; Conditions: trial/context labels	observations x predictors -> zero-inflated response
8	Fit Logistic Decoder	Required: Neural data, Conditions, Time organization Optional: Behavior, Stimuli, Metadata	Conditions: binary labels, left/right, animate/inanimate; Neural: population activity; Time: trial-aligned windows; Behavior: choice labels	observations x neurons -> binary label
9	Fit Multiclass Decoder	Required: Neural data, Conditions, Time organization Optional: Stimuli, Behavior, Metadata	Conditions: multi-class stimulus/task labels; Neural: population activity; Time: aligned observations	observations x neurons -> class label
10	Fit Ridge Decoder	Required: Neural data, Stimuli or Behavior, Time organization Optional: Conditions, Metadata	Stimuli: continuous features/embeddings; Behavior: speed/pose/latent coordinates; Neural: population activity; Time: aligned observations	observations x neurons -> continuous target
11	Fit Temporal Encoding Model	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Time: lags, temporal kernels, stimulus history; Stimuli/Behavior: time-varying covariates; Neural: delayed responses	time x lagged_features -> neural response
12	Fit Temporal Decoder	Required: Neural data, Behavior or Conditions, Time organization Optional: Stimuli, Metadata	Time: neural history/future windows; Behavior: future/past variables; Conditions: upcoming/past labels; Neural: lagged activity	time x lagged_neurons -> target
13	Fit Receptive Field Model	Required: Stimuli, Neural data, Time organization Optional: Conditions, Metadata	Stimuli: spatial/temporal stimulus grid; Neural: spikes/calcium; Time: stimulus preceding response; Metadata: visual area/cell identity	space x time stimulus -> response
14	Fit Spike-Triggered Average	Required: Stimuli, Neural data, Time organization Optional: Conditions, Metadata	Neural: spike/event times; Stimuli: stimulus history; Time: pre-spike window; Conditions: stimulus/task subsets	events x pre-event stimulus window
15	Fit Spike-Triggered Covariance	Required: Stimuli, Neural data, Time organization Optional: Conditions, Metadata	Stimuli: stimulus history ensemble; Neural: spike/event times; Time: pre-event windows	events x stimulus-history vectors
16	Run Cross-Validated Encoding Evaluation	Required: Stimuli or Behavior, Neural data, Time organization Optional: Conditions, Metadata	Time: split definitions, trial/session boundaries; Metadata: leakage groups; Stimuli/Behavior: predictors; Neural: held-out responses	train/test observation splits
17	Run Cross-Validated Decoding Evaluation	Required: Neural data, Conditions or Behavior, Time organization Optional: Stimuli, Metadata	Conditions/Behavior: decoding targets; Neural: held-out activity; Time: split definitions; Metadata: session/group splits	train/test observation splits
18	Compare Feature Spaces for Encoding	Required: Stimuli, Neural data, Time organization Optional: Behavior, Conditions, Metadata	Stimuli: pixel, PCA, ViT, CLIP, semantic features; Neural: response matrix; Time: common alignment; Metadata: same neurons/sessions	multiple feature matrices -> same responses

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
19	Compute Encoding Model Significance	Required: Neural data, Stimuli or Behavior, Time organization Optional: Conditions, Metadata	Validation: null/baseline models; Neural: prediction target; Stimuli/Behavior: predictors; Metadata: multiple-testing groups	model metrics per neuron/population
20	Compute Decoder Confusion Matrix	Required: Neural data, Conditions, Time organization Optional: Stimuli, Behavior, Metadata	Conditions: class labels; Neural: decoder predictions; Time: held-out observations; Metadata: session/area subsets	true label x predicted label

Dimensionality reduction / factor modeling

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
21	Run PCA on Neural Activity	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: population response matrix; Conditions: group/label coloring; Time: observations as trials/time bins; Metadata: region/session subsets	observations x neurons
22	Run Trial-Averaged PCA	Required: Neural data, Conditions, Time organization Optional: Stimuli, Behavior, Metadata	Conditions: stimulus/task labels for averaging; Time: trial structure; Neural: repeated responses	condition x time x neurons or condition x neurons
23	Run demixed PCA	Required: Neural data, Conditions, Time organization Optional: Stimuli, Behavior, Metadata	Conditions: stimulus, choice, time factors, interactions; Behavior: choice/movement covariates; Neural: population activity; Time: trial-aligned time	condition/factor x time x neurons
24	Run Factor Analysis	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: repeated population observations; Time: trials/bins; Conditions: optional grouping; Metadata: neuron/session groups	observations x neurons
25	Run Gaussian Process Factor Analysis	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: spike-count time series; Time: ordered bins/trials; Behavior/Conditions: trajectory annotations	trials x time x neurons
26	Run NMF on Neural Responses	Required: Neural data Optional: Conditions, Stimuli, Time organization, Metadata	Neural: nonnegative responses/counts/events; Conditions/Stimuli: component interpretation labels; Time: observations	observations x neurons, nonnegative
27	Run ICA on Neural Responses	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: mixed population signals; Time: observations/segments; Conditions: optional labels	observations x neurons/signals
28	Run Sparse PCA	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: population matrix; Metadata: sparse neuron loading interpretation; Conditions: labels	observations x neurons
29	Run UMAP Embedding	Required: Neural data Optional: Conditions, Stimuli, Behavior, Time organization, Metadata	Neural: state vectors; Conditions/Stimuli/Behavior: color/target labels; Time: samples/trials	observations x neural features -> 2D/3D
30	Run t-SNE Embedding	Required: Neural data Optional: Conditions, Stimuli, Behavior, Time organization, Metadata	Neural: state vectors; Labels: condition/stimulus/behavior annotations; Time: observations	observations x neural features -> 2D
31	Run Isomap Embedding	Required: Neural data Optional: Conditions, Behavior, Time organization, Metadata	Neural: manifold samples; Time/Behavior: continuous trajectory context; Conditions: labels	observations x neural features -> manifold coords
32	Run Diffusion Map Embedding	Required: Neural data Optional: Conditions, Behavior, Time organization, Metadata	Neural: smooth state manifold; Time: ordered samples; Behavior: slow variables	observations x neural features -> diffusion coords
33	Run CEBRA Embedding	Required: Neural data, Time organization Optional: Behavior, Stimuli, Conditions, Metadata	Behavior: supervised labels/covariates; Stimuli: contrastive labels/features; Time: temporal contrast index; Neural: population activity	samples x neurons with contrastive labels
34	Run LFADS-Style Latent Dynamics Model	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: spike-count sequences; Time: trials x time; Behavior/Conditions: decoding/annotation targets	trials x time x neurons
35	Run Autoencoder Embedding	Required: Neural data Optional: Conditions, Stimuli, Behavior, Time organization, Metadata	Neural: high-dimensional response vectors; Optional labels for interpretation; Time: samples	observations x neurons -> latent

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
36	Run Variational Autoencoder Embedding	Required: Neural data Optional: Conditions, Stimuli, Behavior, Time organization, Metadata	Neural: probabilistic response observations; Optional labels for latent interpretation; Metadata: grouping	observations x neurons -> probabilistic latent
37	Run Contrastive Embedding Model	Required: Neural data, Conditions or Stimuli or Behavior, Time organization Optional: Metadata	Contrast targets: task/stimulus/behavior/time similarity; Neural: samples; Time: positive/negative sampling rules	paired/contrastive sample batches
38	Compute Latent Dimensionality	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: population matrix or fitted latent model; Conditions: subset comparisons; Metadata: area/session grouping	observations x neurons or latent covariance
39	Compute Participation Ratio	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: covariance/spectrum; Conditions: subgroup spectra; Metadata: region/session	observations x neurons -> covariance spectrum
40	Compare Latent Spaces Across Sessions	Required: Neural data, Metadata Optional: Conditions, Stimuli, Behavior, Time organization	Metadata: session/animal/area identity; Neural: latent embeddings; Conditions/Stimuli/Behavior: shared anchors; Time: comparable epochs	session x observations x latent dimensions

Geometry / representational structure

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
41	Compute Representational Similarity Matrix	Required: Neural data Optional: Conditions, Stimuli, Behavior, Time organization, Metadata	Targets: stimuli, conditions, times, behaviors; Neural: response vectors; Time: observations/windows	items x neurons -> items x items similarity
42	Run Representational Similarity Analysis	Required: Neural data, Stimuli or Behavior or Conditions Optional: Time organization, Metadata	Neural geometry vs stimulus/model/behavior geometry; Conditions: labels; Metadata: region/model comparisons	RSM_neural compared to RSM_target
43	Compute Centered Kernel Alignment	Required: Neural data, Stimuli or Behavior or Neural data Optional: Conditions, Time organization, Metadata	Targets: two representation matrices; Stimuli: model features; Neural: region/session matrices; Behavior: covariates	same observations x features in two spaces
44	Compute Canonical Correlation Analysis	Required: Neural data, Neural data or Stimuli or Behavior Optional: Conditions, Time organization, Metadata	Targets: two aligned multivariate spaces; Neural: area A vs B; Stimuli/Behavior: features/covariates; Time: shared observations	observations x features_A and observations x features_B
45	Compute Partial Least Squares Alignment	Required: Neural data, Stimuli or Behavior or Neural data Optional: Conditions, Time organization, Metadata	Predictive alignment between spaces; Stimuli/Behavior: target representation; Neural: source or target	X observations x features -> Y observations x features
46	Run Procrustes Alignment	Required: Neural data, Metadata Optional: Conditions, Stimuli, Behavior, Time organization	Metadata: session/animal/model identity; Conditions/Stimuli: anchor points; Neural: manifold coordinates	matched items x dimensions across spaces
47	Compute Subspace Angles	Required: Neural data Optional: Conditions, Stimuli, Behavior, Metadata	Targets: fitted subspaces by condition/area/session/model; Metadata: grouping; Neural: loadings/bases	basis_A and basis_B
48	Compute Projection Overlap	Required: Neural data Optional: Conditions, Stimuli, Behavior, Metadata	Targets: task/stimulus/condition subspaces; Neural: projection bases; Metadata: area/session groups	basis/projection matrices
49	Compute Neural Manifold Curvature	Required: Neural data Optional: Time organization, Conditions, Behavior, Metadata	Neural: manifold embedding; Time/Behavior: trajectory context; Conditions: subset manifolds	points on learned manifold
50	Compute Geodesic Distances on Neural Manifold	Required: Neural data Optional: Time organization, Conditions, Behavior, Metadata	Neural: embedded manifold graph; Targets: states/stimuli/behavior samples; Time: trajectories	manifold points -> pairwise geodesic matrix
51	Run Principal Geodesic Analysis	Required: Neural data Optional: Conditions, Time organization, Behavior, Metadata	Neural: manifold-valued points; Conditions/Metadata: group comparisons; Time: trajectory points	manifold points + metric

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
52	Run Contrastive PCA	Required: Neural data, Conditions Optional: Stimuli, Behavior, Time organization, Metadata	Conditions: foreground/background groups; Neural: response matrices; Metadata: area/session strata	foreground observations x neurons vs background observations x neurons
53	Run Contrastive Principal Geodesic Analysis	Required: Neural data, Conditions Optional: Behavior, Time organization, Metadata	Conditions: foreground/background manifold groups; Neural: manifold points; Metadata: region/session	foreground/background manifold-valued observations
54	Find Semantic Axes in Neural Space	Required: Neural data, Stimuli or Conditions, Time organization Optional: Behavior, Metadata	Stimuli/Conditions: semantic labels or model concepts; Neural: population responses; Time: aligned observation index	observations x neurons + concept labels/features
55	Project Neural Activity onto Semantic Axis	Required: Neural data, Stimuli or Conditions, Time organization Optional: Behavior, Metadata	Target: known semantic axis such as animate-inanimate; Neural: population activity; Time: aligned observations	observations x neurons -> scalar projection
56	Compute Neural Geometry Stability	Required: Neural data, Time organization Optional: Conditions, Stimuli, Behavior, Metadata	Time: trials/sessions/repeats; Conditions/Stimuli: matched items; Metadata: session/animal/area	repeat/session RSMs or subspaces
57	Compute Population Distance Matrix	Required: Neural data Optional: Conditions, Stimuli, Behavior, Time organization, Metadata	Targets: stimuli, conditions, times, behaviors; Neural: state vectors	items x neurons -> pairwise distance
58	Compute Class Separability in Neural Space	Required: Neural data, Conditions Optional: Stimuli, Behavior, Time organization, Metadata	Conditions: class labels; Neural: population vectors; Metadata: region/session comparisons	observations x neurons + labels
59	Compute Margin Distribution	Required: Neural data, Conditions Optional: Stimuli, Behavior, Time organization, Metadata	Conditions: classifier labels; Neural: classifier scores/projections; Metadata: subsets	observations x classifier scores/margins
60	Map Stimulus Geometry to Neural Geometry	Required: Stimuli, Neural data, Time organization Optional: Conditions, Behavior, Metadata	Stimuli: model/pixel/semantic distances; Neural: response distances; Time: aligned items; Metadata: area/session	stimulus RSM -> neural RSM

Dynamics / state-space

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
61	Fit Linear Dynamical System	Required: Neural data, Time organization Optional: Behavior, Conditions, Metadata	Neural: ordered population states; Time: continuous/trial sequences; Behavior/Conditions: exogenous inputs or labels	trials/time x neurons -> latent states
62	Fit Switching Linear Dynamical System	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Time: ordered sequences; Neural: population states; Conditions/Behavior: regimes/annotations	sequences x time x neurons -> discrete regimes + latent states
63	Fit Hidden Markov Model	Required: Neural data, Time organization Optional: Behavior, Conditions, Metadata	Neural: observation sequences; Time: sequence order; Conditions/Behavior: state interpretation	time x neural observations -> hidden states
64	Fit Autoregressive Neural State Model	Required: Neural data, Time organization Optional: Behavior, Conditions, Metadata	Time: lagged neural history; Neural: continuous/count state vectors; Behavior: exogenous covariates	time x lagged state -> next state
65	Fit Recurrent Neural Network Dynamics Model	Required: Neural data, Time organization Optional: Stimuli, Behavior, Conditions, Metadata	Neural: nonlinear time series; Time: sequences/lags; Stimuli/Behavior: inputs; Conditions: task context	sequence batches: time x features
66	Compute Neural Trajectories	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: latent or population states; Time: ordered trial/epoch paths; Conditions/Behavior: labels along trajectory	trial x time x latent dimensions
67	Align Neural Trajectories Across Trials	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Time: events/warping anchors; Neural: trajectories; Conditions/Behavior: matched trial groups	trials x time x latent dimensions
68	Compute Trajectory Speed	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Time: ordered samples with dt; Neural: latent/population trajectory; Conditions: group comparison	time-indexed trajectory -> speed(t)
69	Compute Trajectory Curvature	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Time: ordered samples; Neural: low-dimensional trajectory; Behavior/Conditions: trajectory annotations	time-indexed trajectory -> curvature(t)
70	Compute Fixed Points of Learned Dynamics	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Input: fitted dynamics model; Neural: state space; Conditions/Behavior: context-dependent fixed points	learned vector field / transition model
71	Analyze Attractor Structure	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: state trajectories; Time: long sequences; Conditions/Behavior: regimes; Metadata: area/session	state-space trajectories + dynamics model

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
72	Compute Rotational Dynamics with jPCA	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: condition-averaged trajectories; Time: aligned event windows; Conditions: movement/task groups	conditions x time x neurons
73	Compute State Transition Matrix	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: discrete latent states or HMM states; Time: ordered state sequence; Conditions/Behavior: context labels	state_t -> state_t+1 counts/probabilities
74	Detect Neural State Changes	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: time series/population states; Time: continuous sequence or trial epochs; Behavior/Conditions: event annotations	time x neural features -> changepoints
75	Detect Neural Events or Motifs	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: repeated temporal patterns; Time: windows/sequences; Behavior/Conditions: motif interpretation	sliding windows x neural features
76	Compare Dynamics Across Conditions	Required: Neural data, Time organization, Conditions Optional: Behavior, Metadata	Conditions: task/stimulus/brain-region groups; Neural: trajectories/dynamics models; Time: aligned epochs	condition x trial x time x neural/latent
77	Estimate Neural Timescales	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: autocorrelation/latent persistence; Time: sampling rate and ordered samples; Metadata: area/session	time series per neuron/population/latent
78	Compute Cross-Area Lagged Influence	Required: Neural data, Time organization, Metadata Optional: Conditions, Behavior	Metadata: brain-area/unit grouping; Neural: area A/B time series; Time: lags; Conditions/Behavior: context subsets	time x features_area_A, time x features_area_B
79	Fit Granger Causality Model	Required: Neural data, Time organization, Metadata Optional: Conditions, Behavior	Metadata: source/target area or unit groups; Time: lag structure; Neural: multivariate time series	time x neural variables with lagged predictors
80	Fit Dynamic Mode Decomposition	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: ordered state snapshots; Time: x_t, x_t+1 pairs; Conditions: subset comparisons	state matrix X and shifted matrix Xprime

Connectivity / interaction

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
81	Compute Noise Correlations	Required: Neural data, Conditions, Time organization Optional: Stimuli, Behavior, Metadata	Conditions/Stimuli: repeated stimulus/task labels; Neural: trial-to-trial residuals; Time: trial windows; Metadata: cell/area grouping	stimulus/condition x trial x neuron
82	Compute Signal Correlations	Required: Neural data, Conditions or Stimuli, Time organization Optional: Metadata	Stimuli/Conditions: tuning labels; Neural: condition-averaged responses; Metadata: neuron/area groups	condition/stimulus x neuron tuning matrix
83	Compute Functional Connectivity Matrix	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: simultaneous unit/signal time series; Time: continuous/trial bins; Metadata: unit/area identities	time/observations x neural units -> unit x unit matrix
84	Compute Partial Correlation Network	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: multivariate activity; Time: observations; Metadata: nodes/areas; Conditions: subset contexts	observations x neural units -> precision/partial corr
85	Fit Graphical Lasso Network	Required: Neural data Optional: Conditions, Time organization, Metadata	Neural: covariance across units; Metadata: graph node identities; Conditions: group-specific networks	observations x neural units -> sparse precision
86	Compute Cross-Correlograms	Required: Neural data, Time organization Optional: Conditions, Metadata	Neural: spike times or event times; Time: precise timestamps/lags; Metadata: unit pairs/areas; Conditions: optional subsets	unit_pair x lag histogram
87	Estimate Coupling GLM	Required: Neural data, Time organization Optional: Stimuli, Behavior, Conditions, Metadata	Neural: target neuron and population history; Stimuli/Behavior: external covariates; Time: lagged coupling filters; Metadata: unit identities	target count ~ stimulus + coupled neural history
88	Compute Population Coupling	Required: Neural data, Time organization Optional: Conditions, Metadata	Neural: unit activity and population activity; Time: aligned observations; Metadata: cell/area groups	observations x neurons -> unit-population coupling
89	Detect Neural Assemblies	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: coactivation patterns; Time: bins/events; Conditions/Behavior: assembly expression contexts; Metadata: cell groups	time/observations x neurons
90	Run Community Detection on Functional Graph	Required: Neural data, Metadata Optional: Conditions, Time organization	Neural: functional connectivity graph/matrix; Metadata: node identities, brain areas; Conditions: graph context	node x node connectivity matrix -> communities

Statistics / validation / visualization

#	Tool	Bundle signature	Target categories and subcategories	Typical materialized view
91	Run Permutation Test	Required: Any target bundle roles used by tested tool Optional: Conditions, Time organization, Metadata	Conditions: labels to shuffle; Time: block/trial constraints; Metadata: grouped permutations; Target: any metric/artifact	observed metric + shuffled null metrics
92	Run Bootstrap Confidence Intervals	Required: Any measured output or source bundle roles Optional: Conditions, Time organization, Metadata	Time: resampling unit definition; Conditions/Metadata: stratified bootstrap; Target: metrics, weights, components	resampled observations/trials/sessions
93	Run Trial Shuffle Control	Required: Neural data, Time organization Optional: Conditions, Stimuli, Behavior, Metadata	Time: trial identity and alignment; Neural: trial responses; Conditions/Stimuli/Behavior: trial labels to break	trial-shuffled observations
94	Run Stimulus Shuffle Control	Required: Stimuli, Neural data, Time organization Optional: Conditions, Metadata	Stimuli: stimulus labels/features; Neural: responses; Time: aligned observations; Conditions: label constraints	stimulus-response mapping with shuffled labels
95	Run Circular Shift Control	Required: Neural data, Time organization Optional: Stimuli, Behavior, Conditions, Metadata	Time: continuous temporal order; Neural: time series; Stimuli/Behavior: shifted covariates; Conditions: block constraints	time series with circular shifts
96	Correct for Multiple Comparisons	Required: Metadata Optional: Neural data, Conditions, Stimuli, Behavior	Metadata: family/grouping of tests; Neural: neuron-wise p-values; Conditions/Stimuli/Behavior: label-wise tests	table of p-values/effects -> corrected q-values
97	Generate Raster Plot	Required: Neural data, Time organization Optional: Conditions, Stimuli, Behavior, Metadata	Neural: spike times/events; Time: trials/events/alignment zero; Conditions/Stimuli/Behavior: row grouping/coloring; Metadata: unit identity	trials x spike/event times
98	Generate PSTH Plot	Required: Neural data, Time organization Optional: Conditions, Stimuli, Behavior, Metadata	Neural: spikes/events/counts; Time: aligned windows/bins; Conditions/Stimuli/Behavior: grouping variables	condition/trial x time bins -> firing rate
99	Generate Latent Trajectory Plot	Required: Neural data, Time organization Optional: Conditions, Behavior, Metadata	Neural: latent trajectories/embeddings; Time: ordered path; Conditions/Behavior: color/annotation labels; Metadata: session/area	trial/condition x time x latent dimensions
100	Generate Structure Discovery Report	Required: Artifacts from any role-signed tools, Metadata Optional: All six roles	Metadata: provenance, assumptions, dataset identity; All roles: inputs/targets of included analyses; Artifacts: models, figures, metrics, nulls	tool artifacts + role manifests -> report

Implementation pattern: RoleBundle -> RoleSignature -> DataJoint provenance

A practical MouseHash implementation can store the role manifest and tool signatures in lightweight tables, while arrays and reports remain disk artifacts. The agent should inspect the RoleBundle, select tools whose RoleSignature is satisfied, ask for missing parameters through BlahML-style prompts, then write the resulting tool run, artifact paths, assumptions, validation checks, and summaries back to the provenance layer.

```
RoleBundle(
conditions=ConditionsTable(...),
stimuli=StimulusTable(...),
behavior=BehaviorTable(...),
neural_data=NeuralDataStore(...),
time=TimeSegmentation(...),
metadata=MetadataStore(...)
)

ToolSignature(
name="fit_ridge_encoding_model",
requires=("stimuli", "neural_data", "time_organization"),
optional=("behavior", "conditions", "metadata"),
produces=("encoding_model", "cv_metrics", "report")
)
```

Source note

Tool names and workflow families are based on the uploaded MouseHash 100 Finding Structure Tools map. The role-bundle target scheme is a design layer added on top so that each tool can be matched against dataset manifests and agent-callable contracts.